

Note 15.4 Double Integrals in Polar Form

Integrals in Polar Coordinates

The double integrals in polar coordinates are:

$$\iint_R f(r, \theta) dA = \int_{\theta=\alpha}^{\theta=\beta} \int_{r=g_1(\theta)}^{r=g_2(\theta)} f(r, \theta) r dr d\theta$$

It can be obtained by the Riemann sum of small pieces of area.

Finding Limits of Integration

Area in Polar Coordinates

The **area** of a closed and bounded region R in the polar coordinates plane is

$$A = \iint_R r dr d\theta$$

Obtain by taking the function as 1.

Changing Cartesian Integral into Polar Integrals

$$\iint_R f(x, y) dx dy = \iint_G f(r \cos \theta, r \sin \theta) r dr d\theta$$

Exercises

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